

Topic: Queue using Singly Linked List

Q. Describe the implementation of a Queue using a Singly Linked List. Detail the logic for the Enqueue and Dequeue operations.

> **Definition to Remember**

> A Q
avoids
**`rear`

Queue Implementation using Singly Linked List

To achieve $O(1)$ time complexity for both Enqueue and Dequeue, we must maintain references to both ends of the list.

| Queue Pointer | Linked List Equivalent | Used For |

| :--- | :

2. Enqueue Operation (Insert at End)

Adds a new element to the rear of the queue.

Algorithm:

```text

### 3. Dequeue Operation (Delete from Beginning)

Removes the element from the front of the queue.

**Algorithm:**

```text

4. Advantages of Linked List Implementation

- **Dynamic Sizing:** Queue can grow infinitely (bounded only by heap memory). No "Queue Full" fixed limit.

- **C

> **Must-Write Points (for 10 marks)**

> 1. Q

> **Quick Recall**

> `Que

new_m